Domain-Adversarial Training Under Label Shift Can Silently Invert Out-of-Distribution Detection

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Abstract

Domain-adversarial training (DANN) is often used to remove site and device shortcuts in medical imaging models, and deployed models are expected to have an out-of-distribution (OOD) monitor that flags unfamiliar inputs. We study how DANN affects such monitors in skin lesion classification, using a lesion-disjoint ISIC 2019 split and smartphone images as the second domain. With gradient reversal, Mahalanobis OOD detection drops from AUROC 0.825 to 0.381 (95% CI $[0.358, 0.404]$) on held-out smartphone images, which then look more in-distribution than the model's own test images. The same happens on three clinical datasets that were not used in training (Fitzpatrick17k 0.405, MED-NODE 0.288, SD-198 0.376), while in-distribution balanced accuracy changes little (paired change -0.009, 95% CI $[-0.026, +0.006]$) and in-distribution ECE stays at 0.20–0.21. Our interventions point to label shift between the two domains as the main driver in this setting. When we change only the class mix that the adversary sees, AUROC on the external datasets changes by 0.24–0.37, and the association remains after adjusting for measured domain leakage. Multi-kernel MMD alignment at similar leakage also inverts detection on the smartphone images, while CORAL does not. DANN moves unfamiliar images toward the majority class. Under label shift, a standard domain probe then shows chance-level leakage, although the domain can still be read within each class. In the same models, the share of melanomas from unseen sources predicted as benign nevus with confidence above 0.9 increases from 13% to 51–81%. Importance-weighted alignment with class priors estimated online from unlabelled target data recovers 66–76% of the AUROC lost in this setting on the smartphone images, Fitzpatrick17k and SD-198 at $\lambda_{\text{adv}} = 0.5$ without loss of in-distribution accuracy, but only 32–45% at $\lambda_{\text{adv}} = 2$, and MED-NODE stays inverted. In a post hoc analysis, with priors from 50 labelled target images the point estimates on the external datasets approach those obtained with the true priors, but in-distribution balanced accuracy drops from 0.48 to 0.39. We recommend checking OOD detection again after domain adaptation and measuring domain leakage within class.

Keywords: domain-adversarial training; out-of-distribution detection; label shift; calibration; skin lesion classification; deployment monitoring

1. Introduction

Skin lesion classifiers often learn acquisition artefacts such as ink markings, rulers, dermatoscope type and contributing site, because these correlate with diagnosis in public datasets (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). A common

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remedy is to make the representation invariant to the acquisition domain, usually with domain-adversarial training and a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016). Deployed models are also expected to have an out-of-distribution (OOD) monitor, which flags inputs that differ from the training data so that a clinician can review them. We study how invariance training affects this monitor.

We find that it can invert the monitor. When the adversarial weight increases, OOD detection does not just get worse but falls below chance, so held-out images from the aligned domain and images from external sources get scores that look more in-distribution than the model's own test images. Standard evaluation does not show this. In-distribution accuracy and calibration change little, and the domain-leakage probe, which the method is supposed to reduce, goes down. In the same models, melanomas from external sources are predicted as benign nevus with confidence above 0.9 four to six times more often than without adaptation; the monitor accepts almost all of these images with or without adaptation, so the increased risk of a missed malignant case comes from the classifier, not from the monitor.

In our experiments, label shift between the aligned domains is the main driver. ISIC 2019 is 50.8% nevus while the smartphone images used as the second domain are 12.3% nevus. When two domains have different class priors, matching their marginal feature distributions forces the model to misplace the classes of one domain (Zhao et al., 2019; Tachet des Combes et al., 2020). We test what this does to OOD monitoring by changing only the class mix that the adversary sees, and we look at where the OOD images end up in feature space.

Contributions.

1. Under a lesion-disjoint split, gradient reversal inverts Mahalanobis OOD detection on held-out smartphone images and on three clinical datasets that were not used in training. In-distribution accuracy and calibration change little. This holds for two training recipes, both within 0.08 balanced accuracy of a standard ERM baseline on the same split.

2. When the class mix seen by the adversary is set to that of the target domain, the inversion on external domains disappears (AUROC difference 0.24–0.37). In a regression that adjusts for measured domain leakage, the label-shift effect remains, so it is not only a matter of deeper invariance.

3. DANN moves images that differ from the source along the domain direction toward the majority class. Under label shift, an unweighted domain probe gives chance-level leakage even though a within-class probe can still read the domain, so we recommend measuring leakage within class.

4. Importance weighting with label-free prior estimates recovers most of the lost AUROC at $\lambda_{\text{adv}} = 0.5$ without loss of in-distribution accuracy, less at $\lambda_{\text{adv}} = 2$, and not on MED-NODE. In a post hoc analysis, priors from 50 labelled target images give point estimates close to those with the true priors, but in-distribution accuracy drops.

2. Related work

Shortcuts and invariance in dermatology. Ink markings shift melanoma predictions (Winkler et al., 2019), classifiers keep competitive ISIC accuracy with the lesion

masked out (Bissoto et al., 2020), and performance degrades on darker skin (Daneshjou et al., 2022). Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) are the usual remedies; our dual-encoder model is a domain separation network and is a vehicle for the intervention, not a contribution.

Invariance under label shift. When class priors differ between domains, matching marginal feature distributions cannot also keep the joint error small (Zhao et al., 2019); importance-weighted alignment (IWDAN, IWCDAN) corrects for this with estimated target class priors (Tachet des Combes et al., 2020), which can be estimated without target labels by black-box shift estimation (Lipton et al., 2018) or EM (Saerens et al., 2002). These results concern classification error. We study what the same misalignment does to OOD monitoring, and show that a standard domain-leakage probe misreports invariance under label shift.

OOD detection after adaptation. Full-spectrum OOD detection (Yang et al., 2023; Zhang et al., 2023) asks that covariate-shifted in-distribution inputs be accepted; our smartphone and clinical images fall in that cell, and Section 5 argues that acceptance is safe only when the model also classifies them. Calibration degrades under shift (Ovadia et al., 2019) and after domain adaptation (Wang et al., 2020). Closest to our observation, Wu et al. report Mahalanobis and ODIN AUROC below 0.5 after adaptation, with rotated MNIST as in-distribution and rotated notMNIST as OOD, with an MMD-based method (DAN) (Wu et al., 2023), as a baseline in a study of continually adaptive detection. We show the inversion on medical images, show that label shift drives it in our setting, and characterise the mechanism.

3. Materials and methods

3.1. Data and splits

ISIC 2019 (Tschandl et al., 2018) (25,331 dermoscopic images, eight classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC) is the in-distribution source. We split it 60/20/20 into train, validation and test *grouped by* `lesion_id` (16,211 / 4,053 / 5,067 images), because 60% of test images in the image-level split share a lesion with training (Tran, 2027). **PAD-UFES-20** (Pacheco et al., 2020) (2,298 smartphone photographs, six of the eight classes) is the domain the adversary aligns against. Its labels are never used in any loss. We split it by patient into `pad_adv` (1,582 images, 961 patients), seen by the adversarial and context branches, and `pad_heldout` (716 images, 412 patients), never seen.

Unseen domains. Four clinical-photograph datasets enter no part of training: Fitzpatrick17k (Groh et al., 2021) (3,887 images obtained), MED-NODE (Giotis et al., 2015) (170 images: 70 melanoma, 100 nevi), SD-198 (Sun et al., 2016) (6,584 images, 198 conditions) and SCIN (Ward et al., 2024) (10,406 images, 5,033 cases; SCIN Data Use License). All four are used for evaluation only and are not redistributed. SD-198 was obtained from the release linked by its first author; it states no licence, and its images originate from DermQuest (Sun et al., 2016). SCIN is mostly inflammatory conditions; only 0.6% of its images map to the ISIC classes, so we use it only as an OOD set for detection and report it separately; no label-based analysis is done on SCIN. Images are cropped to a lesion-centred box and resized to 224×224.

3.2. Model and training

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, z_ℓ^{norm}) used for diagnosis through a linear head, and a context latent $z_c \in \mathbb{R}^{64}$. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}, \tag{1}$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c, $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the latents. AdamW (learning rate 10^{-4}; adversary $\times 30$, classifier $\times 0.2$), 40 epochs, domain-balanced batches. We sweep $\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ with everything else fixed (seeds $\{42, 52, 62, 72, 82\}$ at $\lambda_{\text{adv}} \in \{0, 2\}$, $\{42, \ldots, 46\}$ elsewhere). A second recipe (A4: $z_\ell \in \mathbb{R}^{64}$, classifier $\times 1$, class-weighted loss) serves as a robustness check.

Checkpoints. We report the last checkpoint. Validation balanced accuracy (eight classes, chance 0.125) has a lesion-clustered bootstrap SE of 0.023–0.026, and the best-validation and last checkpoints differ by less than one SE, so best-validation selection picks early epochs by noise. Best-validation values are given for comparison.

Competence. In-distribution balanced accuracy at $\lambda_{\text{adv}} = 0$ is 0.481 (main recipe, last checkpoint) and 0.520 (A4), 0.04–0.08 below a standard single-encoder ERM on the same split (0.558 ± 0.022, five seeds) and about four times chance (0.125). Lesion-disjoint eight-class ISIC 2019 is hard; the image-level split inflates the same model's balanced accuracy by 0.20 (Tran, 2027).

3.3. Interventions on label shift

Class mix seen by the adversary. Keeping the classification loss on all of ISIC, we resample only the ISIC images fed to the adversarial branch so that their class mix is ISIC's own (100%, the default), halfway to PAD's (50%), or PAD's (0%, no label shift between the aligned domains), at $\lambda_{\text{adv}} \in \{0.5, 2\}$ with five seeds (three at 50%).

Importance-weighted alignment. IWDAN and IWCDAN (Tachet des Combes et al., 2020) reweight the source in the adversarial loss by target class priors. In the original method the weights are estimated online during training from unlabelled target predictions; as alternatives we use priors from 50 labelled `pad_adv` images (a few-label setting) and the true `pad_adv` priors (oracle, not deployable). Black-box shift estimates (Lipton et al., 2018) on the final checkpoints serve only as a diagnostic, and an EM estimator (Saerens et al., 2002) failed as implemented and is not used. CDAN (Long et al., 2018) is a class-conditional adversarial baseline.

3.4. Measurements

OOD detection. Class-conditional Mahalanobis with shared covariance (Lee et al., 2018) (primary), k-NN (Sun et al., 2022), maximum cosine to a class centroid, maximum softmax probability (Hendrycks and Gimpel, 2017) and energy (Liu et al., 2020), all fit on ISIC train. AUROC treats OOD as the positive class, so 0.5 is chance and values below 0.5 mean OOD images score as more familiar than ID test images. The five scores are functions of the same latent, so they are not independent evidence.

Table 1: The adversarial-weight sweep (main recipe, lesion-disjoint ISIC split, last checkpoint). Mean ± s.d. over five seeds per λ_{adv} ($\{42, 52, 62, 72, 82\}$ at $\lambda_{\text{adv}} \in \{0, 2\}$, $\{42, \ldots, 46\}$ otherwise). Leakage: linear domain probe on held-out images (chance 0.5). ID: balanced accuracy, eight classes (chance 0.125). Mahalanobis AUROC, OOD positive; k: seeds whose interval lies below 0.5. Best-val: the best-validation checkpoint, for comparison.

λ_{adv}	leakage	ID bal. acc	`pad_heldout`	k	Fitzpatrick17k	k	`pad_heldout` (best-val)
0	0.842 ± 0.050	0.481 ± 0.016	0.825 ± 0.022	0/5	0.712 ± 0.013	0/5	0.808 ± 0.027
0.25	0.502 ± 0.002	0.481 ± 0.010	$\mathbf{0.445 \pm 0.022}$	4/5	$\mathbf{0.396 \pm 0.032}$	5/5	0.525 ± 0.121
0.5	0.508 ± 0.010	0.484 ± 0.005	$\mathbf{0.415 \pm 0.016}$	5/5	$\mathbf{0.377 \pm 0.046}$	5/5	0.426 ± 0.008
1	0.505 ± 0.005	0.482 ± 0.017	$\mathbf{0.390 \pm 0.022}$	5/5	$\mathbf{0.388 \pm 0.039}$	5/5	0.387 ± 0.016
2	0.507 ± 0.009	0.472 ± 0.018	$\mathbf{0.381 \pm 0.023}$	5/5	$\mathbf{0.405 \pm 0.031}$	5/5	0.403 ± 0.028
4	0.527 ± 0.035	0.480 ± 0.016	$\mathbf{0.416 \pm 0.034}$	5/5	$\mathbf{0.354 \pm 0.066}$	5/5	0.433 ± 0.020
8	0.520 ± 0.028	0.480 ± 0.018	$\mathbf{0.439 \pm 0.020}$	5/5	$\mathbf{0.383 \pm 0.036}$	5/5	0.454 ± 0.044

Inversion. We call a cell inverted when the cluster-bootstrap 95% CI of the seed-mean Mahalanobis AUROC lies below 0.5 and at least three seeds' own intervals do (all three in cells with three seeds). Bootstrap draws (2,000) resample ISIC test by lesion and OOD sets by patient (`pad_heldout`), near-duplicate cluster (Fitzpatrick17k) or case (SCIN); MED-NODE and SD-198 carry no patient identifiers and are resampled by image, so their intervals may be too narrow. Runs with in-distribution balanced accuracy below 0.25 are excluded as degraded and reported.

Domain leakage. Balanced accuracy (chance 0.5) of a logistic probe for the source dataset on frozen z_ℓ^{norm}. The probe used in earlier work scores ISIC test against all of PAD, including images the adversary trained on. We therefore also report probes on held-out images only (linear, one-hidden-layer MLP, k-NN), a class-balanced probe that reweights both domains to a common class mix, and a within-class probe averaged over classes.

Calibration and diagnosis. Expected calibration error (15 bins) on ISIC test and on `pad_heldout` (six shared classes), and zero-shot balanced accuracy on `pad_heldout`. Out-of-domain ECE is close to mean confidence minus accuracy here, so these are views of one phenomenon.

4. Results

4.1. OOD detection inverts while in-distribution metrics stay flat

The sweep. At the last checkpoint, Mahalanobis AUROC on `pad_heldout` falls from 0.825 at $\lambda_{\text{adv}} = 0$ to 0.445 at $\lambda_{\text{adv}} = 0.25$ and stays between 0.381 and 0.445 at every larger weight, meeting the inversion criterion at all six non-zero weights; on Fitzpatrick17k it falls from 0.712 to 0.354–0.405 (Table 1). At $\lambda_{\text{adv}} = 2$ the seed-mean interval is $[0.358, 0.404]$ on `pad_heldout` and $[0.392, 0.419]$ on Fitzpatrick17k. In-distribution balanced accuracy stays within 0.472–0.484 (paired $\lambda_{\text{adv}} = 2$ minus $\lambda_{\text{adv}} = 0$: -0.009, 95% CI $[-0.026, +0.006]$) and in-distribution ECE within 0.202–0.210, while out-of-domain ECE on `pad_heldout` rises from 0.305 to 0.724–0.762. All five scores fall below 0.5 on `pad_heldout` at $\lambda_{\text{adv}} = 2$ (k-NN

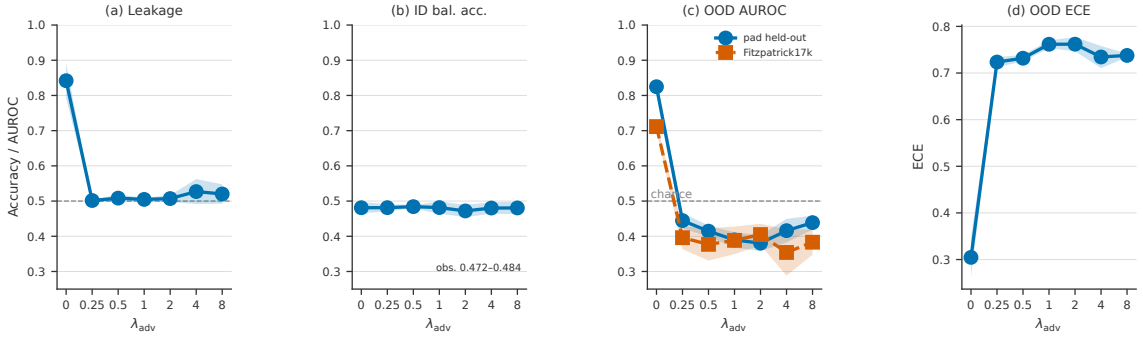


Figure 1: The adversarial-weight sweep at the last checkpoint (main recipe, five seeds; mean ± s.d.). Held-out domain leakage falls to chance and in-distribution balanced accuracy stays flat, while Mahalanobis AUROC on `pad_heldout` and Fitzpatrick17k falls below 0.5 and out-of-domain calibration error rises. Monitoring accuracy and leakage alone would show no problem

0.427, cosine 0.420, MSP 0.455, energy 0.468; best-validation checkpoint); the standard k-th-neighbour k-NN gives the same values within 0.01. The best-validation checkpoint gives the same picture from $\lambda_{\text{adv}} = 0.5$ on, but at $\lambda_{\text{adv}} = 0.25$ it selects an earlier epoch and the inversion is not yet visible (0.525, seed s.d. 0.121; an observation made after the sweep).

Unseen domains. The inversion is not confined to the domain the adversary was trained against (Table 2). On three clinical datasets that entered no part of training, Mahalanobis AUROC falls from 0.712 to 0.405 (Fitzpatrick17k), 0.559 to 0.288 (MED-NODE) and 0.708 to 0.376 (SD-198) at $\lambda_{\text{adv}} = 2$, inverted in all five seeds and at every $\lambda_{\text{adv}} \geq 0.25$ on MED-NODE and SD-198. Clustering near-duplicate SD-198 images (6,016 clusters) leaves every interval and verdict unchanged. On SCIN, whose images are mostly inflammatory conditions outside the ISIC label space, detection falls from 0.723 to 0.551 without inverting.

Other splits. On three further lesion-disjoint splits of ISIC (three seeds each), Mahalanobis AUROC on `pad_heldout` falls from 0.82–0.83 at $\lambda_{\text{adv}} = 0$ to 0.39–0.43 at $\lambda_{\text{adv}} = 2$ (interval below 0.5 in all nine runs), Fitzpatrick17k and SD-198 fall below 0.5 in every split (0.34–0.41), and MED-NODE drops from 0.49–0.57 to 0.24–0.36; in one split MED-NODE is already at chance before adaptation (Appendix K).

A second recipe. With a 64-dimensional latent, unweighted cross-entropy replaced by a class-weighted loss and the last checkpoint (A4), detection again inverts at $\lambda_{\text{adv}} = 0.5$ and $\lambda_{\text{adv}} = 2$ on `pad_heldout` (0.468 $[0.440, 0.495]$; 0.452 $[0.425, 0.478]$) and Fitzpatrick17k (0.444; 0.458), with in-distribution balanced accuracy unchanged (0.520, 0.520, 0.510). On `pad_heldout` all five scores' intervals lie below 0.5; on Fitzpatrick17k k-NN (0.521) and energy (0.516) only collapse. A plain single-encoder DANN collapses to chance (0.51–0.56) without inverting (Appendix A).

Table 2: Unseen domains: Mahalanobis AUROC at $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$ (seed mean, 95% interval at $\lambda_{\text{adv}} = 2$) and the number of seeds whose interval lies below 0.5. Last checkpoint. Intervals: patient (`pad_heldout`), near-duplicate cluster (Fitzpatrick17k, SD-198), case (SCIN), image (MED-NODE).

recipe	set	$\lambda_{\text{adv}} = 0$	$\lambda_{\text{adv}} = 2$	seeds < 0.5	inverted
main	`pad_heldout`	0.825	0.381 $[0.358, 0.404]$	5/5	yes
main	Fitzpatrick17k	0.712	0.405 $[0.392, 0.419]$	5/5	yes
main	MED-NODE	0.559	0.288 $[0.263, 0.315]$	5/5	yes
main	SD-198	0.708	0.376 $[0.362, 0.388]$	5/5	yes
main	SCIN	0.723	0.551 $[0.538, 0.564]$	0/5	no
A4	`pad_heldout`	0.786	0.452 $[0.425, 0.478]$	4/5	yes
A4	Fitzpatrick17k	0.821	0.458 $[0.445, 0.471]$	3/5	yes

4.2. Label shift between the aligned domains drives the inversion in this setting

Intervention. Changing only the class mix of the ISIC images the adversary sees switches the inversion on unseen domains on and off (Table 3). At $\lambda_{\text{adv}} = 2$, with the adversary's ISIC images resampled to PAD's class mix (shift 0%), Mahalanobis AUROC is 0.691 on Fitzpatrick17k, 0.541 on MED-NODE and 0.694 on SD-198, none inverted; with ISIC's own mix (shift 100%) it is 0.340, 0.298 and 0.325, inverted in every seed. Paired by seed, removing the label shift raises AUROC by 0.361, 0.243 and 0.372. The intermediate level falls between the two on Fitzpatrick17k (0.482 at $\lambda_{\text{adv}} = 0.5$, 0.441 at $\lambda_{\text{adv}} = 2$) and SD-198 (0.451, 0.414); on MED-NODE it is already as low as full shift (0.303, 0.299). On `pad_heldout`, the adversary's own target domain, a mild inversion remains without label shift (0.458; paired difference +0.093). One of five shift-0% runs collapsed (in-distribution balanced accuracy 0.139) and is excluded.

Matched leakage. Removing label shift also leaves more domain information in the latent (leakage 0.791 vs 0.634), so the intervention alone does not separate label shift from depth of invariance. A regression of unseen-domain AUROC on label-shift level and leakage, with recipe as a covariate (variance inflation ≈ 1.04), gives label-shift coefficients of -0.166 $[-0.217, -0.115]$ (Fitzpatrick17k), -0.107 $[-0.148, -0.065]$ (MED-NODE) and -0.172 $[-0.221, -0.123]$ (SD-198) for a change from no shift to full shift at fixed leakage (shift coded 0, 0.5, 1; wild cluster bootstrap with Webb weights over recipe $\times$ λ_{adv} $\times$ shift cells, 2,000 draws). On `pad_heldout` the coefficient is $+0.041$ $[-0.033, +0.110]$. Comparing runs at equal leakage across recipes (A4 at shift 100%, leakage 0.78, against the main recipe at shift 0%, leakage 0.79) gives the same direction (+0.23 to +0.26).

A natural example without DANN. Label shift alone can invert a class-conditional Mahalanobis score. Training on BCN and treating HAM as OOD, Mahalanobis AUROC is already 0.369 $[0.336, 0.401]$ at $\lambda_{\text{adv}} = 0$, while k-NN gives 0.658. HAM-heldout is 67% nevus against 31% in BCN test; reweighting HAM to BCN's class mix gives 0.489 $[0.453, 0.524]$,

Table 3: Label-shift intervention: Mahalanobis AUROC (main recipe, last checkpoint) by the class mix the adversary sees. Shift 100%: ISIC's own mix; 0%: PAD's mix. Leakage: linear probe on ISIC test against all of PAD, as in earlier work. Shift 0% at $\lambda_{\text{adv}} = 2$: four seeds after excluding one degraded run (in-distribution balanced accuracy $0.139 < 0.25$); shift 50%: three seeds. These runs are trained separately from the sweep with the same recipe and seeds, so shift 100% differs from Table 1 by run-to-run variation (e.g. 0.368 vs 0.381 on `pad_heldout` at $\lambda_{\text{adv}} = 2$).

λ_{adv}	shift	leakage	`pad_heldout`	Fitzpatrick17k	MED-NODE	SD-198
0	–	0.928	0.825	0.712	0.559	0.708
0.5	0%	0.783	0.498	0.672	0.525	0.664
0.5	50%	0.629	0.490	0.482	0.303	0.451
0.5	100%	0.628	0.361	0.385	0.310	0.365
2	0%	0.791	0.458	0.691	0.541	0.694
2	50%	0.658	0.454	0.441	0.299	0.414
2	100%	0.634	0.368	0.340	0.298	0.325

and no class-matched AUROC differs from 0.5 (post hoc). In the main setting the OOD sets are *less* nevus-heavy than ISIC, so detection works at $\lambda_{\text{adv}} = 0$; DANN creates the condition by moving them toward nevus.

Other alignment objectives. The inversion is not specific to gradient reversal. Replacing the adversary with a multi-kernel MMD penalty (Long et al., 2015) on the same model, at held-out leakage matched to gradient reversal (0.506 vs 0.502) and with in-distribution accuracy within 0.04, inverts detection on `pad_heldout` (0.354 ± 0.022; all five scores 0.34–0.43), as Wu et al. (2023) observed with rotated MNIST and notMNIST. On the unseen datasets it only collapses (Fitzpatrick17k 0.553, MED-NODE 0.523, SD-198 0.535). CORAL alignment at the same leakage leaves detection intact (0.820; 0.788, 0.702, 0.788). Matching the full feature distribution, adversarially or by kernel MMD, is what misaligns classes under label shift; matching second moments is not enough.

4.3. Mechanism: misaligned classes behind chance-level leakage

Unfamiliar images move to the majority class. At $\lambda_{\text{adv}} = 2$ with full label shift, the share of `pad_heldout` images whose nearest class centroid is nevus rises from 0.28 to 0.63, and to 0.75–0.85 on the unseen sets; without label shift it stays at 0.12 and 0.39–0.49. Images of non-nevus classes are pulled into the nevus region: split by true class, non-nevus OOD images fall to AUROC 0.42 (main recipe) and 0.47 (A4), whereas images still nearest a non-nevus centroid stay at 0.62 and 0.64. Let AUROC_c compare all OOD images with the ID test images of class c. Because AUROC averages over ID samples, the pooled AUROC equals $\sum_c \pi_c \, \text{AUROC}_c$, with π_c the share of class c in the ID test set (verified numerically to 10^{-15}). Nevus is half of the ISIC test set, so OOD images placed in the nevus region are enough to push the pooled score below chance.

The displacement follows the domain direction. For each OOD image we measured how far it lies along the direction separating ISIC from PAD in the $\lambda_{\text{adv}} = 0$ model, and how much closer to its nearest ID centroid it moves at $\lambda_{\text{adv}} = 2$. The two correlate on every lesion dataset (Spearman 0.33–0.47), while the deviation orthogonal to that direction does not predict the pull ($|\rho| \leq 0.07$ on four of five sets). Between two $\lambda_{\text{adv}} = 0$ models of different seeds the same correlation is smaller (0.11–0.29); the DANN-specific excess exceeds 0.15 on all four unseen datasets (0.18–0.33) but not on `pad_heldout` (0.057), and is larger in the pre-projection backbone (0.24–0.45 on all five).

Chance-level leakage hides the misalignment. The domain probe of Table 3 (as in earlier work) is scored partly on images the adversary trained on, which is why it reads 0.928 at $\lambda_{\text{adv}} = 0$ against 0.842 for the held-out probe of Table 1. On held-out images a linear probe reads the domain at chance from $\lambda_{\text{adv}} = 0.5$ on (0.502–0.506), but a k-NN probe still reads it (0.55–0.61), and under label shift an unweighted probe misreports invariance (Table 4). With full label shift DANN reaches 0.510 on an unweighted held-out probe, yet within each class the domain remains decodable (0.618), more so than in any other condition. Correcting the class mix the adversary aligns (shift 0%, or importance weighting with the true priors) raises the unweighted reading (0.63–0.64) while lowering the within-class one (0.56–0.57). Under label shift, DANN matches the marginal distributions by misaligning classes, as the theory of Zhao et al. (2019) implies; leakage should therefore be measured within class (post hoc).

4.4. Safety-relevant misclassification and mitigation

Melanomas called benign. With a threshold accepting 95% of ISIC validation images, 0.90–1.00 of the malignant images from external datasets are accepted at both $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$, so the monitor does not catch them in either case. What changes is the classifier: the share of melanomas predicted nevus with confidence above 0.9 rises from 12.5% [7.5, 17.6] to 51.1% [41.4, 61.0] on Fitzpatrick17k ($n = 83$) and from 12.9% [8.8, 17.2] to 81.4% [75.9, 86.5] on SD-198 ($n = 113$) between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$ (Appendix H); BCC, SCC and AK follow the same pattern (50–87% at $\lambda_{\text{adv}} = 2$). Removing the label shift seen by the adversary (shift 0%) lowers the melanoma share at $\lambda_{\text{adv}} = 2$ from 53.0% to 21.4% on Fitzpatrick17k and from 83.2% to 33.8% on SD-198 (retrained runs of Table 3).

No reliable scoring-side fix among the variants tested. Reversing the sign of the score is not a remedy: at $\lambda_{\text{adv}} \geq 0.5$ it reaches only 0.55–0.63 on `pad_heldout` and Fitzpatrick17k with a false-positive rate of 0.84–0.98 at 95% true-positive rate across datasets, and at $\lambda_{\text{adv}} = 0$ it would turn 0.808 into 0.192. Relative Mahalanobis (Ren et al., 2021) and per-class standardised distances stay below 0.5 on all three unseen datasets at $\lambda_{\text{adv}} = 2$.

Importance-weighted alignment. IWDAN reweights the source in the adversarial loss by estimated target class priors, and how well it works depends on the estimate. With weights updated online from unlabelled target predictions, as in the original method, the implied target priors end 0.43–0.55 (total variation; mean 0.515) from the truth: nevus is down-weighted from 0.51 toward 0.23–0.28, against a true 0.10. IWDAN then keeps leakage at the DANN level (0.642 vs 0.634) and recovers 32% of the lost AUROC on `pad_heldout` (0.513) and 45% on Fitzpatrick17k (0.508), while MED-NODE (0.350) and SD-198 (0.488)

Table 4: Domain leakage on held-out images under three probes, and unseen-domain detection ($\lambda_{\text{adv}} = 2$, main recipe, last checkpoint). Unweighted: as usually reported; class-balanced: both domains reweighted to a common class mix; within-class: class-size-weighted mean of per-class probes. Fitz: Mahalanobis AUROC on Fitzpatrick17k.

condition	unweighted	class-balanced	within-class	Fitz
DANN, shift 100%	0.510	0.602	0.618	0.340
DANN, shift 0%	0.629	0.539	0.560	0.691
IWDAN (online)	0.534	0.562	0.597	0.508
IWDAN (50 labels)	0.653	0.545	0.583	0.741
IWDAN (true priors)	0.645	0.549	0.576	0.717

stay at or below chance. At $\lambda_{\text{adv}} = 0.5$ the online priors are closer to the truth (total variation 0.38–0.47, mean 0.428) and the same method recovers more: 0.667 on `pad_heldout` and 0.633 on Fitzpatrick17k (66% and 76% of the gap), at leakage 0.665 against 0.628 for DANN, and SD-198 recovers to 0.607 (71%); MED-NODE improves from 0.310 to 0.447 but stays inverted. Unlike the variants with known priors, this costs no in-distribution accuracy (0.482 against 0.483), and the within-class probe is lower than under DANN (0.588 against 0.618). *EM* estimation was unstable: in two of five seeds it placed mass on classes absent from PAD, so we treat it as not having run correctly. With priors from 50 *labelled target images* (total variation 0.10–0.14), detection on the unseen domains returns to the level obtained with the true priors (Fitzpatrick17k 0.741 vs 0.717, SD-198 0.734 vs 0.693, MED-NODE 0.606 vs 0.563; `pad_heldout` 0.619 vs 0.647), and within-class leakage is lower than under DANN (0.58 vs 0.62). This is a promising but post hoc mitigation: neither the 50-label nor the true-prior variant meets our leakage criterion, because unweighted leakage rises (0.81 and 0.78 against 0.63), and both cost in-distribution balanced accuracy (0.39–0.40, against 0.48 both under DANN and before adaptation). Read descriptively, detection improves as the prior error falls (0.52: partial; 0.10–0.14: oracle level; 0: oracle), but our planned rank-correlation test across estimators was not supported (Spearman -0.33, $[-0.80, +0.28]$), with EM contributing invalid priors. As a diagnostic, black-box shift estimates computed on the final checkpoints were further from the truth (0.65–0.76): their assumption of unchanged class-conditional predictions fails across these domains, with target BCC and AK images predicted as MEL or NV. Label-free prior estimates thus remain about 0.5 total variation off and recover detection only in part; 50 labels bring the error to 0.10–0.14 and detection to the oracle level. Conditional alignment does not help: CDAN (0.378 on `pad_heldout`, 0.420 on Fitzpatrick17k) and IWCDAN (0.360, 0.384) stay inverted at $\lambda_{\text{adv}} = 2$ (Appendix F).

5. Discussion

Accepting shifted inputs. Full-spectrum OOD detection argues that covariate-shifted inputs should be accepted (Yang et al., 2023). After DANN, our models accept images from external sources more readily than their own test images, but their zero-shot balanced

accuracy on these images is only 0.244 at $\lambda_{\text{adv}} = 2$ (chance 0.167). AUROC below 0.5 means that the ranking is reversed: the monitor scores the shifted images as more typical than real in-distribution data. Whether a monitor tolerates shifted inputs is a separate question, answered by the acceptance rate at the deployed threshold (0.90–1.00 here, Appendix H) together with the accuracy on the same images. We think claims of tolerance to covariate shift should be reported together with the accuracy on the same inputs.

Why label shift matters. Zhao et al. (2019) showed that when class priors differ, matching marginal feature distributions cannot keep the classes of both domains in place. Our results show the effect of this on a distance-based monitor. OOD images that differ from the source along the domain direction are moved into the region of the majority class, and a class-conditional score then treats them as typical. The BCN→HAM case shows that an OOD set with more nevi than the ID set can invert such a score even without DANN. In the main setting the OOD sets have fewer nevi than ISIC, and DANN creates the problem by moving them toward nevus. Without label shift the inversion is only seen on the adversary's target domain, where a smaller within-class pull remains. This misplacement is also why the unweighted domain probe reaches chance level, since it measures marginal rather than class-conditional alignment.

Checkpoint selection. We noticed after the sweep that at $\lambda_{\text{adv}} = 0.25$ the last checkpoint is inverted on both `pad_heldout` and Fitzpatrick17k, while best-validation selection picks earlier epochs where the seeds still disagree (seed s.d. 0.121). Validation accuracy does not separate these checkpoints, so a model selected in the usual way may look safe only because training was stopped early.

Fixes at scoring time. Reversing the score only helps if one already knows that the model is inverted. At $\lambda_{\text{adv}} = 0$ it would break a working monitor, and at $\lambda_{\text{adv}} \geq 0.5$ most OOD images are still accepted at a 95% ID-acceptance threshold. Relative Mahalanobis, which is less sensitive to class structure, also stays below chance. Among the variants we tried, only changes during training, through the class priors used by the adversary, improved detection.

Recommendations. After a domain-alignment step, we recommend that practitioners (i) check OOD detection again on images from unseen sources, per class and pooled, (ii) report domain leakage within class and not only unweighted, (iii) compare the class priors of the aligned domains and correct for them if they differ, which in our case required a small labelled sample from the target domain, and (iv) report results for more than one checkpoint.

6. Limitations

Scope. The inversion is shown in dermatology only, and its spread to unseen domains is shown for gradient reversal; multi-kernel MMD inverts detection on the target domain but only collapses it on the unseen datasets. We attempted two other modalities and neither experiment took place: on Camelyon17 no adversarial setting we tried removed hospital information while preserving the classifier, and on chest radiographs the detector was near chance on the OOD set before any adaptation (AUROC 0.522), so there was

nothing to invert. **Architecture.** The main results use a dual-encoder model; the A4 recipe changes the latent size and loss, but a plain single-encoder DANN collapses detection to chance without inverting, and whether deeper invariance would invert it is untested: with a stronger adversary (warm-up, higher adversary learning rate, deeper head) its held-out leakage stayed at 0.70, above our target of 0.55, and detection stayed near chance (0.52 on `pad_heldout`, 0.62 on Fitzpatrick17k). Without label shift, raising λ_{adv} to 8 brought held-out leakage to chance in only one of five runs; in that run detection on the unseen datasets did not invert, which is consistent with label shift being necessary but rests on a single model. **Competence.** In-distribution balanced accuracy is 0.04–0.08 below a standard ERM on the same lesion-disjoint split. **Statistics.** MED-NODE and SD-198 carry no patient identifiers and are resampled by image, so their intervals may be too narrow; SCIN is used only as an OOD set because almost none of it maps to the ISIC classes. The five OOD scores are functions of one latent and are not independent evidence. **Mechanism.** A two-domain Gaussian toy model with label shift did not reproduce the inversion, so the conditions we identify may not be sufficient in general. **Warning signals.** No statistic computed on in-distribution data alone flagged the inverted runs; the share of deployment images predicted as the majority class tracked inversion (Spearman -0.77 and -0.75 over 64 runs) but needs unlabelled target data. **Mitigation.** The few-label remedy is a post hoc analysis, costs in-distribution accuracy, and did not meet our leakage criterion. **Data.** We obtained 3,887 of 16,577 Fitzpatrick17k images.

7. Conclusion

We showed that domain-adversarial training can make an OOD monitor rank held-out images from the aligned domain and from external sources as more in-distribution than the model's own test images, while in-distribution accuracy changes little. In our experiments (one dermatology task, one main architecture), this is driven mainly by label shift between the domains that the adversary aligns. The adversary moves unfamiliar images toward the majority class, and the usual leakage probe does not detect this misalignment. Checking OOD detection on unseen sources after adaptation, measuring leakage within class and correcting for class priors are cheap steps. With a small labelled sample from the target domain, the last of these also reduced the problem in a post hoc analysis.

Use of AI tools

[PENDING: Check the MIDL 2027 policy on AI use and adapt.] We used a large language model (Claude, Anthropic) to help plan analyses, review code and analysis reports, and draft and edit the manuscript text. All experiments were run by the authors, all reported numbers were checked against the analysis outputs with an automated number checker, and the authors take full responsibility for the content.

Data and code availability

https://github.com/TranDucVinh-20225538/CSG-SKin. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Other alignment objectives and the single-encoder model

All values are Mahalanobis, k-NN, cosine, MSP and energy AUROC on `pad_heldout` (image-level ISIC split, best-validation checkpoint, patient-clustered intervals), so they are comparable with each other but not with the main tables.

Table 5: Single-encoder DANN (EfficientNet-B3, one linear head and a domain head; five seeds at $\lambda_{\text{adv}} \in \{0, 2\}$, three elsewhere). Detection collapses to about chance from $\lambda_{\text{adv}} = 0.25$ and does not invert.

λ_{adv}	Mahalanobis	k-NN	cosine	MSP	energy
0	0.700 [0.683, 0.717]	0.935 [0.924, 0.944]	0.875 [0.861, 0.887]	0.857 [0.844, 0.868]	0.901 [0.891, 0.910]
0.25	0.539 [0.514, 0.564]	0.548 [0.526, 0.571]	0.530 [0.508, 0.553]	0.548 [0.529, 0.567]	0.536 [0.515, 0.555]
0.5	0.525 [0.499, 0.549]	0.539 [0.516, 0.561]	0.519 [0.498, 0.542]	0.543 [0.522, 0.563]	0.543 [0.524, 0.561]
1	0.555 [0.530, 0.581]	0.541 [0.519, 0.562]	0.517 [0.495, 0.539]	0.517 [0.497, 0.537]	0.498 [0.479, 0.516]
2	0.521 [0.494, 0.546]	0.542 [0.520, 0.562]	0.514 [0.491, 0.535]	0.540 [0.522, 0.559]	0.529 [0.511, 0.547]
4	0.512 [0.486, 0.538]	0.514 [0.489, 0.539]	0.490 [0.467, 0.512]	0.519 [0.498, 0.540]	0.501 [0.481, 0.521]
8	0.525 [0.502, 0.549]	0.558 [0.536, 0.580]	0.526 [0.505, 0.545]	0.566 [0.545, 0.585]	0.551 [0.531, 0.570]

MMD at weight 10,000 reaches leakage 0.836 only by reducing in-distribution balanced accuracy to 0.337, so it never reaches the invariance gradient reversal attains with the classifier intact. On the dual-encoder model at matched held-out leakage, multi-kernel

Table 6: Non-adversarial objectives on the single-encoder model, by weight (three seeds). None inverts detection; MMD lowers it while keeping every score above 0.6.

objective	weight	Mahalanobis	k-NN	cosine	MSP	energy
ERM	–	0.834 [0.819, 0.849]	0.940 [0.929, 0.949]	0.888 [0.874, 0.900]	0.826 [0.813, 0.838]	0.823 [0.809, 0.836]
CORAL	1	0.910 [0.899, 0.920]	0.970 [0.962, 0.977]	0.959 [0.949, 0.967]	0.827 [0.814, 0.839]	0.834 [0.821, 0.847]
CORAL	100	0.998 [0.996, 1.000]	0.998 [0.996, 1.000]	0.999 [0.996, 1.000]	0.944 [0.937, 0.951]	0.629 [0.614, 0.644]
MMD	1	0.811 [0.791, 0.827]	0.930 [0.919, 0.940]	0.853 [0.837, 0.866]	0.815 [0.800, 0.828]	0.804 [0.788, 0.819]
MMD	100	0.724 [0.696, 0.751]	0.791 [0.768, 0.812]	0.668 [0.644, 0.690]	0.729 [0.713, 0.745]	0.663 [0.643, 0.684]
MMD	10000	0.664 [0.650, 0.678]	0.740 [0.725, 0.755]	0.633 [0.617, 0.650]	0.670 [0.650, 0.689]	0.696 [0.678, 0.713]
IRM	1	0.918 [0.905, 0.930]	0.951 [0.940, 0.959]	0.878 [0.865, 0.889]	0.521 [0.498, 0.543]	0.464 [0.440, 0.486]
IRM	100	0.724 [0.713, 0.736]	0.752 [0.738, 0.765]	0.702 [0.690, 0.715]	0.496 [0.482, 0.509]	0.542 [0.528, 0.556]
GroupDRO	1	0.860 [0.843, 0.875]	0.941 [0.930, 0.949]	0.842 [0.828, 0.856]	0.646 [0.627, 0.664]	0.581 [0.558, 0.603]

MMD does invert detection on `pad_heldout` (Section 4.2). IRM's low MSP and energy values reflect a classifier that has collapsed (balanced accuracy 0.167 at weight 100), not an inversion of the feature scores.

Appendix B. Label-shift intervention: full results

Main recipe, last checkpoint; one degraded shift-0% run (in-distribution balanced accuracy 0.139) is excluded. k/n: seeds whose interval lies below 0.5.

The regression uses one point per run ($n = 75$): Mahalanobis AUROC on label-shift level (coded 0, 0.5, 1), unweighted leakage and recipe. Intervals come from a wild cluster bootstrap with Webb weights over recipe $\times$ λ_{adv} $\times$ shift cells (2,000 draws).

Appendix C. BCN→HAM: label shift without DANN (post hoc)

ISIC 2019 contains two sites, Hospital Clínic Barcelona (BCN) and HAM10000 (HAM), both dermoscopic. Training on BCN with a lesion-disjoint held-out part of HAM as the OOD set, Mahalanobis AUROC is 0.369 $[0.336, 0.401]$ at $\lambda_{\text{adv}} = 0$, below chance in all three seeds, while k-NN gives 0.658 $[0.627, 0.688]$. The reverse direction (HAM→BCN) gives 0.846.

Reweighting HAM to BCN's class mix recovers 92% of the distance to 0.5, no class-matched interval excludes 0.5, and dispersion ratios are close to one. The inversion is therefore a label-shift effect on the pooled score; we cannot exclude a small within-class contribution.

Appendix D. Domain-direction analysis

For each OOD image, the domain score is its projection on a linear ISIC-versus-`pad_adv` classifier fitted on the $\lambda_{\text{adv}} = 0$ model's features; the pull is the decrease in its Mahalanobis distance to the nearest ID centroid, normalised by the median ID distance of each run, from $\lambda_{\text{adv}} = 0$ to $\lambda_{\text{adv}} = 2$ (matched seeds). The orthogonal control uses the magnitude of the deviation orthogonal to the domain direction. The placebo replaces the $\lambda_{\text{adv}} = 2$ model with a $\lambda_{\text{adv}} = 0$ model of a different seed.

Across λ_{adv} (cross-seed design), the correlation is already present at $\lambda_{\text{adv}} = 0.25$ and is slightly larger at $\lambda_{\text{adv}} = 2$ on four of five sets (differences 0.02–0.08). On the two BCN/HAM

Table 7: Mahalanobis AUROC by the class mix seen by the adversary.

set	shift	λ_{adv}	mean ± s.d.	95% CI	k/n	inverted
pad_heldout	100%	0	0.825 ± 0.022	[0.810, 0.839]	0/5	not met
pad_heldout	100%	0.5	0.361 ± 0.029	[0.338, 0.383]	5/5	met
pad_heldout	50%	0.5	0.490 ± 0.058	[0.466, 0.512]	2/3	not met
pad_heldout	0%	0.5	0.498 ± 0.025	[0.478, 0.517]	0/5	not met
pad_heldout	100%	2	0.368 ± 0.017	[0.344, 0.391]	5/5	met
pad_heldout	50%	2	0.454 ± 0.009	[0.432, 0.476]	3/3	met
pad_heldout	0%	2	0.458 ± 0.003	[0.438, 0.477]	4/4	met
Fitzpatrick17k	100%	0	0.712 ± 0.013	[0.698, 0.725]	0/5	not met
Fitzpatrick17k	100%	0.5	0.385 ± 0.023	[0.373, 0.398]	5/5	met
Fitzpatrick17k	50%	0.5	0.482 ± 0.143	[0.469, 0.494]	2/3	not met
Fitzpatrick17k	0%	0.5	0.672 ± 0.046	[0.661, 0.684]	0/5	not met
Fitzpatrick17k	100%	2	0.340 ± 0.065	[0.328, 0.352]	5/5	met
Fitzpatrick17k	50%	2	0.441 ± 0.076	[0.428, 0.454]	2/3	not met
Fitzpatrick17k	0%	2	0.691 ± 0.031	[0.680, 0.702]	0/4	not met
MED-NODE	100%	0	0.559 ± 0.019	[0.531, 0.587]	0/5	not met
MED-NODE	100%	0.5	0.310 ± 0.055	[0.284, 0.337]	5/5	met
MED-NODE	50%	0.5	0.303 ± 0.054	[0.272, 0.336]	3/3	met
MED-NODE	0%	0.5	0.525 ± 0.050	[0.497, 0.554]	1/5	not met
MED-NODE	100%	2	0.298 ± 0.032	[0.271, 0.326]	5/5	met
MED-NODE	50%	2	0.299 ± 0.052	[0.269, 0.331]	3/3	met
MED-NODE	0%	2	0.541 ± 0.027	[0.512, 0.572]	0/4	not met
SD-198	100%	0	0.708 ± 0.019	[0.695, 0.722]	0/5	not met
SD-198	100%	0.5	0.365 ± 0.029	[0.353, 0.377]	5/5	met
SD-198	50%	0.5	0.451 ± 0.133	[0.439, 0.461]	2/3	not met
SD-198	0%	0.5	0.664 ± 0.058	[0.653, 0.675]	0/5	not met
SD-198	100%	2	0.325 ± 0.069	[0.315, 0.336]	5/5	met
SD-198	50%	2	0.414 ± 0.047	[0.402, 0.425]	3/3	met
SD-198	0%	2	0.694 ± 0.031	[0.683, 0.704]	0/4	not met
SCIN	100%	0	0.723 ± 0.018	[0.710, 0.736]	0/5	not met
SCIN	100%	0.5	0.534 ± 0.015	[0.521, 0.547]	0/5	not met
SCIN	50%	0.5	0.586 ± 0.101	[0.575, 0.599]	1/3	not met
SCIN	0%	0.5	0.718 ± 0.036	[0.708, 0.729]	0/5	not met
SCIN	100%	2	0.519 ± 0.073	[0.507, 0.532]	1/5	not met
SCIN	50%	2	0.566 ± 0.051	[0.553, 0.578]	0/3	not met
SCIN	0%	2	0.732 ± 0.035	[0.722, 0.742]	0/4	not met

Table 8: Label-shift coefficient at fixed leakage.

set	n	coefficient	95% CI	excludes 0	VIF (leakage)	VIF (recipe)	identifiable
Fitzpatrick17k	75	-0.166	[-0.217, -0.115]	yes	1.04	1.03	yes
MED-NODE	75	-0.107	[-0.148, -0.065]	yes	1.04	1.03	yes
SD-198	75	-0.172	[-0.221, -0.123]	yes	1.04	1.03	yes
pad_heldout (not counted)	75	0.041	[-0.033, 0.110]	no	1.04	1.03	yes

directions, the orthogonal deviation also predicts the pull, so the pattern is specific to the cross-modality setting.

Appendix E. Held-out domain probes

In the A4 recipe, the held-out linear and MLP probes still read the domain at $\lambda_{\text{adv}} \in \{0.5, 2\}$ (0.66–0.71) although detection inverts, so inversion does not require chance-level leakage. At $\lambda_{\text{adv}} = 0.25$ in the main recipe (observation after the fact, $n = 5$), the one seed whose

Table 9: BCN→HAM at $\lambda_{\text{adv}} = 0$ (three seeds, lesion-clustered intervals). Class-matched: HAM images of class c against BCN test images of class c. Dispersion: median squared Mahalanobis distance to the image's own class centroid, HAM over BCN.

class	BCN test share	HAM share	class-matched Mahalanobis	class-matched k-NN	dispersion ratio
MEL	0.222	0.118	0.443 [0.362, 0.527]	0.620 [0.540, 0.693]	0.92
NV	0.306	0.667	0.483 [0.418, 0.546]	0.639 [0.584, 0.693]	0.97
BCC	0.215	0.052	0.498 [0.437, 0.561]	0.656 [0.585, 0.729]	0.99
AK	0.077	0.011	0.685 [0.570, 0.792]	0.595 [0.447, 0.733]	1.20
BKL	0.109	0.108	0.526 [0.445, 0.615]	0.623 [0.542, 0.705]	1.01
DF	0.021	0.012	0.577 [0.360, 0.768]	0.717 [0.593, 0.852]	1.05
SCC	0.042	0.020	0.407 [0.275, 0.528]	0.714 [0.594, 0.826]	0.90
pooled, unweighted			0.369 [0.337, 0.402]	0.658 [0.626, 0.688]	
pooled, HAM reweighted to BCN mix			0.489 [0.453, 0.524]		

Table 10: Spearman correlation between domain score and pull (latent z_ℓ^{norm}), and the excess over the seed placebo (paired bootstrap). Backbone: pre-projection features.

set	ρ_{dom}	ρ_{orth}	placebo ρ_{dom}	excess (latent)	excess (backbone)
`pad_heldout`	+0.344	−0.068	+0.288	+0.057 [+0.007, +0.105]	+0.331 [+0.283, +0.374]
Fitzpatrick17k	+0.419	−0.066	+0.205	+0.214 [+0.196, +0.232]	+0.383 [+0.364, +0.400]
MED-NODE	+0.433	−0.144	+0.106	+0.327 [+0.243, +0.407]	+0.349 [+0.256, +0.432]
SD-198	+0.467	−0.030	+0.203	+0.264 [+0.250, +0.278]	+0.449 [+0.436, +0.461]
SCIN	+0.333	+0.021	+0.150	+0.183 [+0.172, +0.193]	+0.239 [+0.227, +0.251]

held-out probe still reads the domain (0.67) does not invert on any set, three of the four seeds at chance invert on all four, and one inverts only on MED-NODE.

Appendix F. Importance-weighted alignment by prior estimate

EM estimation (Saerens et al., 2002) failed as implemented, including after a bug fix: in two of five seeds it placed mass on DF and VASC, which are absent from PAD, and its results are not used in any comparison. Black-box shift estimates computed on the final IWDAN checkpoints are a diagnostic, not a separate method (TV 0.65–0.76). Conditional variants stay inverted (last checkpoint, five seeds unless noted): CDAN 0.378 / 0.420 (`pad_heldout` / Fitzpatrick17k) at $\lambda_{\text{adv}} = 2$ and 0.372 / 0.403 at $\lambda_{\text{adv}} = 0.5$; IWCDAN 0.360 / 0.384 at $\lambda_{\text{adv}} = 2$ and 0.389 / 0.410 at $\lambda_{\text{adv}} = 0.5$ At $\lambda_{\text{adv}} = 0.5$, IWDAN (online) reaches 0.667 / 0.633 and IWDAN with the true priors 0.680 / 0.742.

Appendix G. Sign reversal and scoring-side variants

Reversing the Mahalanobis score with the direction chosen on `pad_heldout` at $\lambda_{\text{adv}} = 2$ gives 0.192 at $\lambda_{\text{adv}} = 0$ and 0.546–0.625 on `pad_heldout` and Fitzpatrick17k at $\lambda_{\text{adv}} \geq 0.5$; on MED-NODE it reaches 0.69–0.71; false-positive rates at 95% true-positive rate stay at 0.84–0.98 across datasets. Relative Mahalanobis at $\lambda_{\text{adv}} = 2$: Fitzpatrick17k 0.443, MED-

Table 11: Domain probes (main recipe, best-validation checkpoint, five seeds). The first column includes PAD images the adversary trained on.

λ_{adv}	all of PAD (earlier work)	held-out linear	held-out MLP	held-out k-NN
0	0.927	0.827	0.921	0.935
0.25	0.614	0.538	0.547	0.586
0.5	0.581	0.506	0.500	0.548
1	0.563	0.504	0.500	0.560
2	0.555	0.502	0.500	0.578
4	0.636	0.516	0.500	0.570
8	0.663	0.550	0.523	0.608

Table 12: IWDAN at $\lambda_{\text{adv}} = 2$ (main recipe, last checkpoint, five seeds) by how the target class priors are obtained. TV: total variation from the true `pad_adv` priors. Leakage: unweighted linear probe on ISIC test against all of PAD; within-class: held-out within-class probe.

priors	TV	ID bal. acc	leakage	within-class	`pad`	Fitz	MED	SD
DANN (no weighting)	–	0.48	0.634	0.618	0.368	0.340	0.298	0.325
IWDAN (online)	0.43–0.55	0.48	0.642	0.596	0.513	0.508	0.350	0.488
IWDAN (50 labels)	0.10–0.14	0.39	0.806	0.583	0.619	0.741	0.606	0.734
IWDAN (true priors)	0	0.40	0.780	0.576	0.647	0.717	0.563	0.693
$\lambda_{\text{adv}} = 0.5$								
DANN (no weighting)	–	0.48	0.628	0.618	0.361	0.385	0.310	0.365
IWDAN (online)	0.38–0.47	0.48	0.665	0.588	0.667	0.633	0.447	0.607
IWDAN (true priors)	0	0.40	0.760	0.562	0.680	0.742	0.622	0.741

NODE 0.376, SD-198 0.427, `pad_heldout` 0.439; per-class standardised Mahalanobis: 0.478, 0.340, 0.447, 0.424.

Appendix H. Clinical consequence

Threshold: 95% acceptance of ISIC validation images, set per model. Label map: Fitzpatrick17k melanoma, basal cell carcinoma, squamous cell carcinoma and actinic keratosis to MEL, BCC, SCC, AK; SD-198 Malignant_Melanoma and Lentigo_Maligna_Melanoma to MEL, Basal_Cell_Carcinoma to BCC, Bowen's disease and Keratoacanthoma to SCC, and Actinic_solar_Damage (actinic keratosis) to AK; other SD-198 classes are excluded. `pad_heldout` has nine melanomas and is not used.

Appendix I. Negative results

Warning signals. Thresholds set at the 95th percentile of each statistic over $\lambda_{\text{adv}} = 0$ runs. A centroid-collapse statistic computed on in-distribution data only did not track inversion (sensitivity 0.15–0.38; Spearman with `pad_heldout` AUROC 0.03 and -0.14 for the validation and test versions, over 64 runs). The share of unlabelled deployment images predicted as the majority class tracked it (Spearman -0.77 $[-0.85, -0.63]$ against `pad_heldout` AUROC and -0.75 $[-0.85, -0.57]$ against Fitzpatrick17k AUROC over 64 runs; sensitivity

Table 13: Malignant images from external datasets (main recipe, last checkpoint, five seeds; bootstrap 95% intervals): share accepted by the monitor at the 95% ID threshold, and share predicted NV with confidence above 0.9.

			accepted		predicted NV, conf. > 0.9	
dataset	class	n	$\lambda_{\text{adv}} = 0$	$\lambda_{\text{adv}} = 2$	$\lambda_{\text{adv}} = 0$	$\lambda_{\text{adv}} = 2$
Fitzpatrick17k	MEL	83	0.961 [0.938, 0.981]	0.990 [0.978, 1.000]	0.125 [0.075, 0.176]	0.511 [0.414, 0.610]
Fitzpatrick17k	BCC	257	0.932 [0.915, 0.948]	0.985 [0.977, 0.992]	0.068 [0.047, 0.092]	0.742 [0.702, 0.781]
Fitzpatrick17k	SCC	212	0.896 [0.870, 0.921]	0.946 [0.925, 0.965]	0.069 [0.048, 0.093]	0.498 [0.446, 0.551]
Fitzpatrick17k	AK	24	0.958 [0.925, 0.992]	1.000 [1.000, 1.000]	0.092 [0.033, 0.167]	0.725 [0.583, 0.858]
SD-198	MEL	113	0.963 [0.940, 0.981]	0.979 [0.958, 0.996]	0.129 [0.088, 0.172]	0.814 [0.759, 0.865]
SD-198	BCC	60	0.933 [0.903, 0.960]	0.983 [0.967, 0.997]	0.043 [0.013, 0.077]	0.857 [0.793, 0.913]
SD-198	SCC	119	0.921 [0.887, 0.950]	0.976 [0.958, 0.992]	0.089 [0.061, 0.121]	0.721 [0.659, 0.778]
SD-198	AK	60	0.950 [0.927, 0.973]	0.997 [0.990, 1.000]	0.033 [0.013, 0.053]	0.867 [0.813, 0.917]

1.00, specificity 0.75 on twelve non-inverting runs with $\lambda_{\text{adv}} > 0$ under the Fitzpatrick17k label, 0.33 on six under the `pad_heldout` label). **Toy model.** Two-domain Gaussian mixtures with a majority class at 50% in the source and a label-shift level from 0 to 1, with a small encoder and gradient reversal, produced no inversion at any setting (Mahalanobis AUROC 0.57–0.66); leakage in the toy stayed near 0.5–0.58, so it may not reproduce the conditions of the real data.

Appendix J. Experiments that did not take place

Camelyon17 by hospital. In the domain-adversarial runs with hospital-disjoint held-out slides, the selected checkpoints were all from the warm-up epoch before gradient reversal took effect (15 of 15 runs), so these runs do not test the hypothesis. In runs whose selected checkpoint followed the warm-up (epochs 6–10), the hospital probe fell by at most 0.044 with in-distribution accuracy ≥ 0.99, and at $\lambda_{\text{adv}} = 300$ the probe fell by 0.081 while out-of-domain accuracy dropped from 0.91 to 0.70. In no setting did DANN remove hospital information while preserving the classifier. **Chest radiographs.** With NIH ChestX-ray14 and pediatric radiographs as the aligned domains, Mahalanobis AUROC on Shenzhen was 0.522 before any adaptation, below our headroom requirement, so we stopped before training adversarial models.

Appendix K. Other lesion-disjoint splits

Appendix L. Image-level split (lesion leakage; for reference only)

The following results use the image-level ISIC split common in published work, in which 60% of test images share a lesion with training. They overstate in-distribution accuracy by about 0.20 and must not be compared with the main tables; we include them because earlier versions of this analysis used them.

Table 14: Three further lesion-disjoint ISIC splits (main recipe, last checkpoint, three seeds each). ID and leakage: mean ± s.d.; AUROC: seed mean with pooled 95% interval (lesion groups rebuilt per split; patient, cluster or image resampling as in the main tables). No run met the degraded-model criterion. At $\lambda_{\text{adv}} = 2$ every interval lies below 0.5 and all three seeds' intervals do in every cell. Leakage: linear probe on held-out images.

split	λ_{adv}	ID bal. acc	leakage	`pad_heldout`	Fitzpatrick17k	MED-NODE	SD-198
7	0	0.443 ± 0.008	0.845 ± 0.034	0.835 [0.820, 0.848]	0.713 [0.699, 0.726]	0.491 [0.457, 0.525]	0.724 [0.712, 0.737]
7	2	0.440 ± 0.008	0.509 ± 0.007	0.406 [0.383, 0.430]	0.344 [0.330, 0.358]	0.245 [0.220, 0.271]	0.338 [0.327, 0.349]
123	0	0.449 ± 0.011	0.840 ± 0.028	0.822 [0.807, 0.835]	0.739 [0.725, 0.751]	0.569 [0.535, 0.600]	0.745 [0.733, 0.757]
123	2	0.452 ± 0.025	0.512 ± 0.010	0.431 [0.407, 0.454]	0.408 [0.395, 0.422]	0.324 [0.294, 0.354]	0.392 [0.380, 0.404]
2027	0	0.460 ± 0.023	0.831 ± 0.039	0.827 [0.813, 0.841]	0.715 [0.701, 0.729]	0.544 [0.512, 0.575]	0.706 [0.693, 0.720]
2027	2	0.482 ± 0.007	0.514 ± 0.014	0.393 [0.369, 0.415]	0.376 [0.362, 0.390]	0.357 [0.326, 0.385]	0.348 [0.337, 0.360]

Table 15: Image-level split (lesion leakage), best-validation checkpoint; leakage is the probe of earlier work. With the two non-adversarial controls. Five seeds at $\lambda_{\text{adv}} \in \{0, 0.25, 1, 2, 8\}$, three at $\lambda_{\text{adv}} \in \{0.5, 4\}$.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.843 ± 0.023	0.764 ± 0.038	0.291 ± 0.016
0.25	5	0.571 ± 0.027	0.692	0.100	0.694	0.509 ± 0.023	0.460 ± 0.057	0.295 ± 0.028
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	5	0.591 ± 0.035	0.695	0.097	0.735	0.457 ± 0.026	0.414 ± 0.086	0.258 ± 0.012
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.427 ± 0.025	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391